

Day – 5 Config the OSPF Practical.

Task – 1 Create three routers in two OSPF areas in cisco.

- Router 1 (R1): Located entirely inside Area 1.
- Router 2 (R2): Acts as an Area Border Router , connecting Area 1 and Area 0.
- Router 3 (R3): Located entirely inside Area 0 (Backbone).
- IP Addressing Plan
- Link between R1 and R2 (Area 1): 192.168.12.0/30 (Mask: 255.255.255.252)
- R1 (GigabitEthernet0/0): 192.168.12.1
- R2 (GigabitEthernet0/0): 192.168.12.2
- Link between R2 and R3 (Area 0): 192.168.23.0/30 (Mask: 255.255.255.252)
- R2 (GigabitEthernet0/1): 192.168.23.1
- R3 (GigabitEthernet0/0): 192.168.23.2
-
- Step 1: Configure Router 1 (Area 1)
- Open R1, go to the CLI, and run the following commands:
- enable
- configure terminal
- hostname R1
- interface GigabitEthernet0/0
- ip address 192.168.12.1 255.255.255.252
- no shutdown
- exit
- router ospf 1
- network 192.168.12.0 0.0.0.3 area 1
- End
-
- Step 2: Configure Router 2 (ABR - Area 1 & Area 0)
- Open R2, go to the CLI, and run the following commands:
- enable
- configure terminal
- hostname R2
- interface GigabitEthernet0/0
- ip address 192.168.12.2 255.255.255.252
- no shutdown
- exit

- interface GigabitEthernet0/1
- ip address 192.168.23.1 255.255.255.252
- no shutdown
- exit
- router ospf 1
- network 192.168.12.0 0.0.0.3 area 1
- network 192.168.23.0 0.0.0.3 area 0
- End
-
- Step 3: Configure Router 3 (Area 0)
- Open R3, go to the CLI, and run the following commands:
- enable
- configure terminal
- hostname R3
- interface GigabitEthernet0/0
- ip address 192.168.23.2 255.255.255.252
- no shutdown
- exit
- router ospf 1
- network 192.168.23.0 0.0.0.3 area 0
- end

Task – 2 Set OSPF area 0 as the backbone of topology.

- 1. Router 1 (Area 1 Only) :- Open R1 and check its neighbors. Because R1 is entirely inside Area 1, it will only form an adjacency with R2.
- R1# show ip ospf neighbor
- Neighbor ID Pri State Dead Time Address Interface
- 192.168.12.2 1 FULL/BDR 00:00:32 192.168.12.2 GigabitEthernet0/0
- Observation: R1 is successfully adjacent with the ABR (R2) in Area 1.
-
- 2. Router 2 (ABR - Area 0 & Area 1):- Open R2 and check its neighbors. As the ABR, R2 bridges both worlds and will show neighbors in separate areas.
- R2# show ip ospf neighbor
- Neighbor ID Pri State Dead Time Address Interface
- 192.168.12.1 1 FULL/DR 00:00:35 192.168.12.1 GigabitEthernet0/0
- 192.168.23.2 1 FULL/BDR 00:00:33 192.168.23.2 GigabitEthernet0/1
- Observation: R2 maintains a FULL adjacency with R1 (via interface connected to Area 1) and R3 (via interface connected to Area 0 backbone).

- 3. Router 3 (Area 0 Backbone Only):- Open R3 and check its neighbors. R3 resides in the backbone area and connects directly to R2.
- R3# show ip ospf neighbor
- Neighbor ID Pri State Dead Time Address Interface
- 192.168.23.1 1 FULL/DR 00:00:34 192.168.23.1 GigabitEthernet0/0
- Observation: R3 is successfully adjacent with the ABR (R2) in Area 0.

Task - 3 Connect two pc in OSPF network.

- Step 1: Expand the Topology with PCs and New Subnets
- Area 1 LAN (192.168.1.0/24): Connect PC1 to R1 using a Copper Straight-Through cable.
- Area 0 LAN (192.168.3.0/24): Connect PC3 to R3 using a Copper Straight-Through cable.
- IP Addressing Plan for the PCs:
- PC1 (Area 1):
- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1 (R1's interface)
- PC3 (Area 0):
- IP Address: 192.168.3.10
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.3.1 (R3's interface)
- Step 2: Configure Router Interfaces and OSPF Updates
-
- 1. Configure Router 1 (R1 - Area 1)
- Open R1, go to the CLI, and add the new LAN interface and OSPF network statement:
- enable
- configure terminal
- interface GigabitEthernet0/1
- ip address 192.168.1.1 255.255.255.0
- no shutdown
- exit
- router ospf 1
- network 192.168.1.0 0.0.0.255 area 1
- End
-
- 2. Configure Router 3 (R3 - Area 0)
- Open R3, go to the CLI, and add its LAN interface and OSPF network statement:

- enable
- configure terminal
- interface GigabitEthernet0/1
- ip address 192.168.3.1 255.255.255.0
- no shutdown
- exit
- router ospf 1
- network 192.168.3.0 0.0.0.255 area 0
- End
-
- Step 3: Test Connectivity via Ping

Task – 4 Simulate the router by shut down.

- Step 1: Simulate the Link Failure:- Open R2, go to configuration mode, and shut down the interface facing Area 0:
 - enable
 - configure terminal
 - interface GigabitEthernet0/1
 - shutdown
 - End
- Step 2: Observe the Impact (Failure State):- Check OSPF Neighbors: Run show ip ospf neighbor on R2.
 - The neighbor in Area 0 (R3) will disappear from the list after the OSPF Dead Timer expires (default is 40 seconds on Ethernet segments).
 - Check the Routing Table: Run show ip route on R1 (in Area 1).
 - The inter-area routes pointing to Area 0 (192.168.3.0/24 and 192.168.23.0/30) will completely disappear from R1's routing table because R2 can no longer reach the backbone.
 - Test Ping: If you try to ping from PC1 (Area 1) to PC3 (Area 0), the requests will time out (Request timed out).
 -
- Step 3: Restore the Link (Recovery)
 - Go back to R2, bring the interface back up, and watch how quickly OSPF recovers:
 - configure terminal
 - interface GigabitEthernet0/1
 - no shutdown
 - end

